

Q-eHACK 2026

Real-time Voting and Counting System

Hackathon Document

Submitted by:

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1 Abstract

This project presents a Secure Real-Time Voting System using **QNX** RTOS. The system ensures secure voting through authentication, prevents duplicate voting, and provides real-time result updates using multi-threading, IPC, and mutex mechanisms.

2 Problem Statement

Traditional voting systems suffer from:

- Lack of security
- Duplicate voting
- No real-time monitoring
- Risk of vote loss

A secure and real-time solution is required.

3 Objective

- Build a secure voting system
- Ensure real-time vote counting
- Prevent duplicate votes
- Use RTOS concepts

4 System Architecture

The system uses multi-threading where votes are sent using IPC to a high-priority counter task. Mutex ensures safe shared data access.

- Counter Task
- Timer Task
- IPC (MsgSend / MsgReceive)
- Named Pipes

5 Working / Flow

1. Voter enters ID
2. System verifies voter
3. Face verification
4. Candidate selection
5. Vote sent via IPC
6. Counter updates result
7. Exit verification

6 Security Features

- Duplicate vote prevention
- Face verification
- Admin authentication
- Fraud detection
- Rate limiting

7 RTOS Concepts Used

- Multithreading
- Priority scheduling
- Mutex
- IPC
- Timer control

8 Output

The system produces the following outputs demonstrating its functionality:

- **Live Dashboard:** Displays real-time voting statistics
- **Vote Confirmation:** Shows successful vote casting
- **Fraud Detection:** Detects duplicate voting attempts
- **Rate Limiting:** Temporarily locks system for abnormal voting
- **Final Results:** Displays winner and vote count

9 Output Screenshots

9.1 Admin Menu

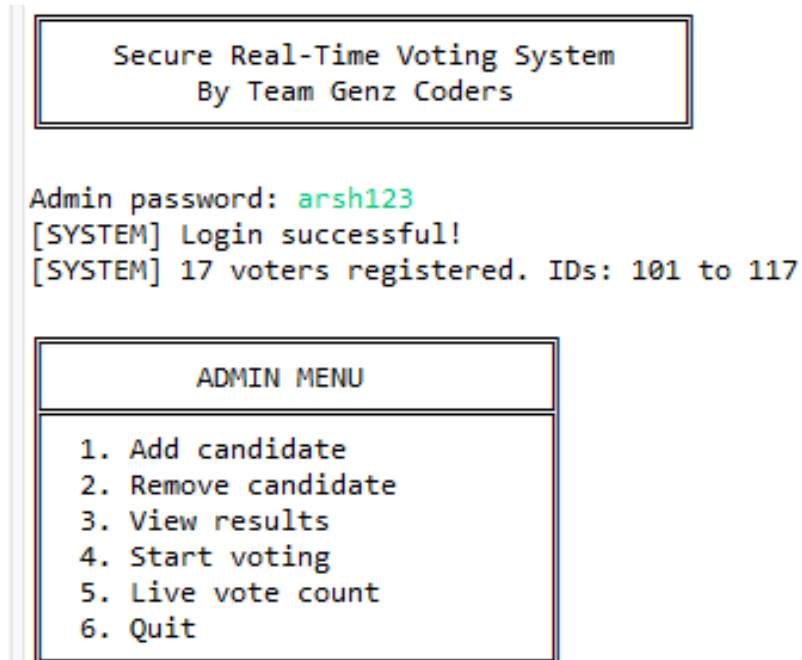


Figure 1: Admin menu interface

9.2 Vote Casting and Verification

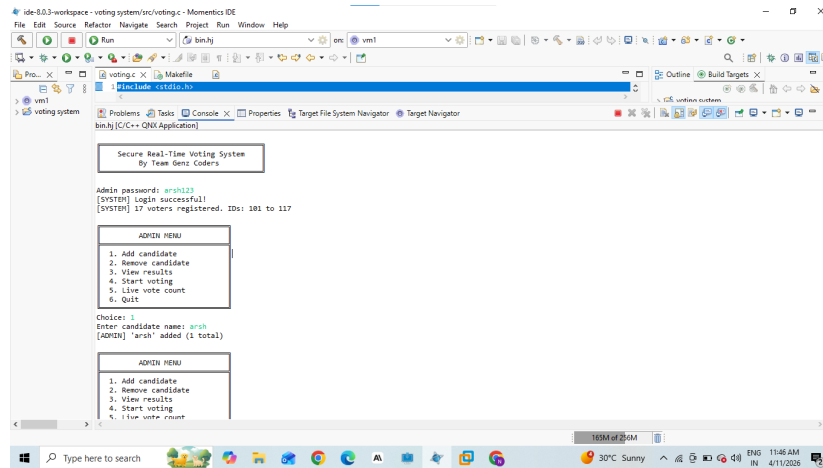


Figure 2: Vote casting and verification

9.3 Fraud Detection

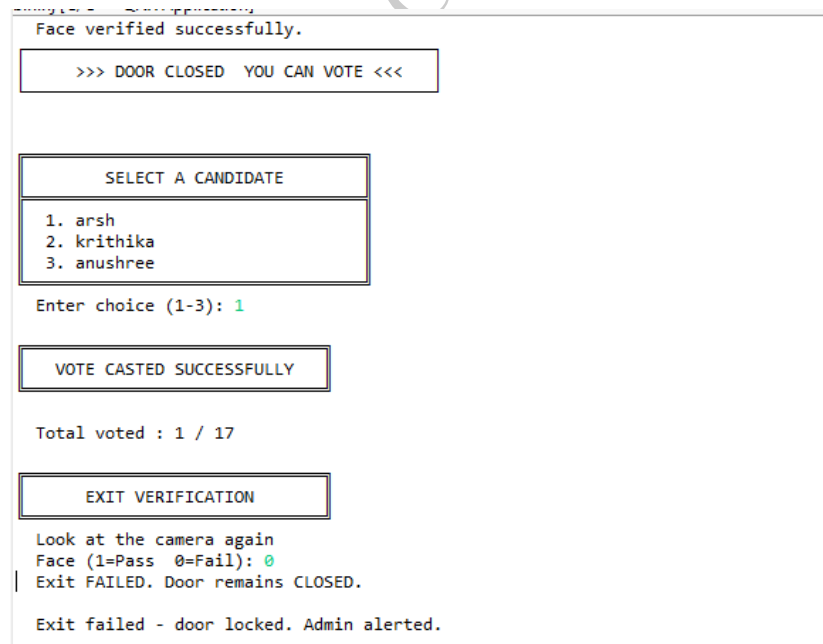


Figure 3: Duplicate vote detection

9.4 Rate Limiting Lock

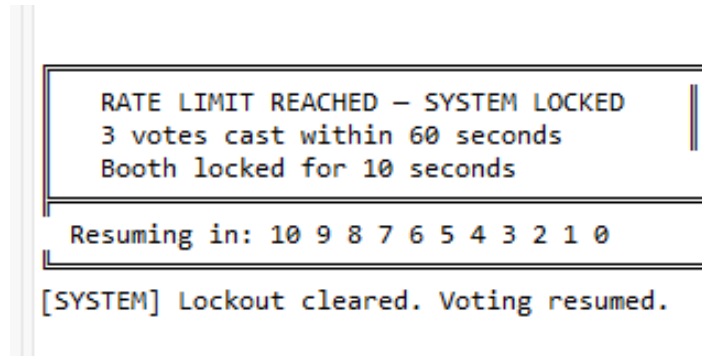


Figure 4: System lock due to excessive attempts

9.5 Final Result

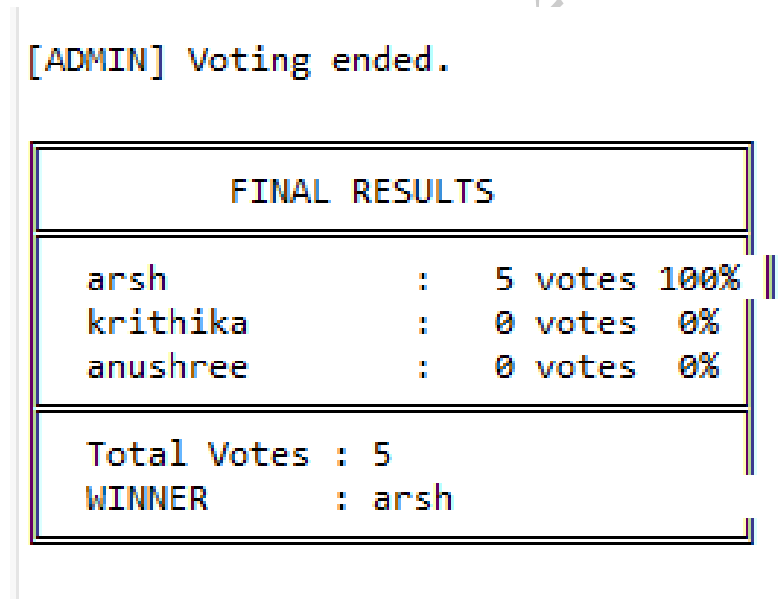


Figure 5: Final vote results and winner

10 How to Run the Code

10.1 Requirements

- QNX Momentics IDE



- **VMware**

10.2 Steps

1. Open **QNX** IDE
2. Create C project
3. Paste code in **main.c**
4. Build using **QNX** compiler
5. Run on **QNX** system

10.3 Execution

- Enter admin password
- Add candidates
- Start voting
- Cast votes
- View results

11 Conclusion

A secure and real-time voting system was successfully implemented using **QNX** RTOS concepts.